CHRISTOPHER NASER

[Phone] | [Email]
[LinkedIn] | [GitHub] | [Portfolio]

COMPUTER SCIENCE | PRE-MED | AI AND MEDICINE

PROFILE AND PURPOSE

Computer Science junior and pre-med student at Wayne State University with a 4.0 GPA and recognized achievement in chemistry. Driven to build a career where medicine and artificial intelligence strengthen one another, with interests in deep learning, medical imaging, visual-spatial pattern recognition, biomedical data, and clinical reasoning. My goal is to bridge the gap between AI and medicine by applying computational thinking to meaningful biomedical and clinical problems while remaining grounded in scientific rigor, patient care, and accessibility.

SELECTED RESEARCH AND SCIENTIFIC WORK

Detroit Road Quality and Driver Outcomes Research (2026)

Independent quantitative research | more than 70 hours of dedicated research and documentation

- Independently completed research design, data preparation, statistical modeling, visualization, interpretation, and documentation in Python, producing a research report of more than 20 pages from roadway records and original survey data.
- Applied regression ANOVA, multiple regression, confidence intervals, hypothesis testing, decision trees, model diagnostics, and classification metrics to study crash safety, route choice, and driver well-being.
- Found no significant linear relationship between pavement rating and controlled crash or injury outcomes, while behavioral analyses showed stronger links between road conditions, route avoidance, mental fatigue, and driving stress.

Cancer Disparities in Detroit and Metro Detroit (2024)

10-page secondary medical and public-health research paper

- Synthesized peer-reviewed literature on colorectal, breast, prostate, and lung cancer disparities in Detroit, focusing on survival, late-stage diagnosis, treatment access, neighborhood deprivation, comorbidity, and prevention.
- Found that many apparent racial differences in cancer survival narrowed after socioeconomic and neighborhood factors were considered, pointing to structural disadvantage, delayed care, and limited treatment access as major contributors rather than race alone.
- Connected Detroit property-tax overassessment and financial instability to reduced socioeconomic opportunity and healthcare access, then evaluated responses including community health education, broader screening access, and outreach programs for underserved communities.

The Case of K'Jon: A Moral Argument (2025)

Clinical ethics review | Ethical Issues in Health Care

- Evaluated a Detroit birth-trauma case involving a delayed emergency C-section through non-maleficence, fidelity, justice, patient communication, equitable care, and professional accountability, emphasizing preventable harm and the long-term importance of trust in medicine.

EDUCATION

Wayne State University

B.S. Computer Science, Pre-Med
Junior | Expected 2028
GPA: 4.0

Selected science: Organic/General Chemistry I-II with labs; Cellular Biology; Anatomy & Physiology with lab; Calculus-Based Physics I-II; Calculus I-III; Probability & Statistics

AWARDS AND HONORS

Harold B. Cutter Memorial Award
Organic Chemistry, Wayne State University, 2026

2026 Chair's Honor List
Department of Chemistry, Wayne State University

CORE SKILLS

- **Programming & software:** Python, Java, C++; JavaFX; object-oriented design
- **Statistics & modeling:** Regression/ANOVA; confidence intervals; hypothesis testing with t, chi-square, and F distributions; decision trees; model diagnostics
- **Data & research:** pandas; Matplotlib; data preparation/visualization; survey design; primary/secondary research; literature synthesis; scientific writing
- **Medicine & science:** Bioethics; cancer disparities; health equity; clinical case analysis

ORGANIC CHEMISTRY LABORATORY EXPERIENCE

15+ laboratory experiments | Organic Chemistry I and II

- **Core techniques:** Titration, recrystallization, simple and fractional distillation, reflux, vacuum filtration, and melting-point analysis.
- **Separation & characterization:** TLC/chromatography, Rf and eluent selection, IR spectroscopy, H-NMR interpretation, and purity assessment.
- **Synthesis & quantitative analysis:** Organic reaction setup, green-chemistry methods, percent yield/recovery, product isolation, and reaction interpretation.
- **Representative work:** Ferrocene/acetylferrocene TLC, terpene IR analysis, acetanilide bromination/recrystallization, and 2-methylcyclohexanol dehydration by reflux/distillation.

TEACHING, SERVICE AND COMMUNICATION

TotemTutor - Founder and Chemistry Educator
Founded and built a free chemistry education channel, handling lesson planning, visual explanations, recording, editing, publishing, and tutoring. The work combines sustained preparation, clear scientific communication, educational entrepreneurship, and a commitment to accessible chemistry instruction.

Detroit Tutoring Project - Incoming Tutor
Beginning Fall 2026, will tutor K-2 students in Detroit, adapting explanations to each learner's pace, strengths, and needs while building confidence and trust. The role emphasizes patience, empathy, flexibility, and individualized care.

wikiHow - Instructional Author
Published a step-by-step guide on selling shoes to a consignment shop, turning practical experience into a process readers can repeat independently. The article required audience awareness, procedural teaching, concise organization, and clear written communication.

SELECTED COMPUTING PROJECTS

Advanced Cash Register - C++
Engineered a complete transaction system around custom linked structures, customer queues, inventory persistence, search and sorting algorithms, templates, operator overloading, inheritance, and polymorphism. The design required careful software architecture, data-structure decisions, validation, and end-to-end state management.

Client-Side Chess - JavaFX
Built an interactive chess application with board rendering, move validation, turn management, rule-based game-state logic, and event-driven controls, requiring algorithmic reasoning, object-oriented modeling, state tracking, and responsive GUI design.

Text Analysis and Similarity - Java
Developed a modular text-analysis pipeline for tokenization, vocabulary richness, word frequency, sentiment classification, report generation, and cosine similarity. The application connects text processing with vector mathematics and interpretable analytical outputs.

Address Book - JavaFX
Created a desktop contact manager with input validation, search and filtering, sorting, birthday and city queries, and file persistence, emphasizing user-centered interface design, data integrity, structured records, and reliable local storage.